

Shared evidence | Museum investigation

Evolution: build the evidence case

Name _____ Date _____

Use this page with one response route. F and A are scientific summaries; D contains constructed sample data for fictional species.

F | Fossils

Pakicetus: about 50 million years ago, four walking legs. Ambulocetus: about 50-48 million years ago, large feet suited to movement in water. Dorudon: about 40-33 million years ago, flippers and very small hind legs. Approximate ages and broad features: Natural History Museum.

D | Compare DNA

Constructed sample data: the same aligned 100 DNA positions were compared in fictional species. Q/A differ at 2 positions; Q/B at 11; Q/C at 24. Scientists normally examine much more DNA and combine evidence.

A | Anatomical pattern

A human forelimb and a five-digit whale flipper share corresponding regions: one upper bone, two lower bones, wrist bones and digits. Their shapes and functions differ. This shared arrangement is called homologous.

L | Evidence limits

Fossilisation and discovery are incomplete. Similar body outlines can arise independently. One short DNA region is a sample. No card identifies an exact direct ancestor of a living species.

W1. Repair: “Whales came from living humans because both have similar finger-bone patterns.”

W2. Repair: “Q and A are identical. The graph proves A became Q.”

W3. Why combine different evidence types?

Supported route | Build a museum caption

Keep the shared evidence page beside you. You may point, annotate or dictate.
Words: shared ancestor • corresponding bones • fossil • sampled DNA • supports.

S1. Choose Pakicetus and Dorudon. Name one limb difference and say which is older.

S2. Complete: the human limb and whale flipper share This supports

Compared with Q	Differences in 100 positions
A	2
B	11
C	24

S3. Constructed sample data above: which is most similar to Q? Use a number.

S4. Write a short visitor caption using one fossil fact and the limb pattern. Add one thing the evidence cannot tell us.

Standard route | Make a claim worth displaying

Produce a visitor panel using the shared evidence. A drawing with an explanation is welcome.

M1. Compare Pakicetus and Dorudon using approximate age and one limb feature. Explain why this supports change over time.

M2. Explain why the corresponding human and whale limb bones support shared ancestry. Do not claim one living species became the other.

M3. Constructed sample data: Q/A = 2, Q/B = 11, Q/C = 24 differences in the same 100 positions. Identify the closest pair involving Q and justify it.

M4. Finish your panel with one specific limit and one additional piece of evidence you would seek.

Stretch route | Review the curator's reasoning

Critique the claims, then give the museum a better evidence statement.

Claim 1: "The three fossils prove a complete, direct chain to every modern whale."

Claim 2: "Q and A must have identical whole genomes because only two differences were counted."

T1. Identify one error in each claim. Write corrected versions.

T2. Constructed sample data: a second aligned DNA region has $Q/A = 4$, $Q/B = 14$ and $Q/C = 25$ differences per 100 positions. Does it strengthen the same relationship? Explain, including a limitation.

T3. Design the next check: choose a new DNA region, a new fossil or a new anatomical feature. State a prediction and how the result could change confidence.

T4. Write a two-sentence visitor caption using at least two evidence types without claiming certainty about an exact ancestor.

Exit and reflection | Evidence before confidence

Complete independently; short answers or a recorded oral explanation are enough.

E1. Constructed DNA sample: Q/A = 2, Q/B = 11, Q/C = 24 differences per 100 positions. Which comparison supports the closest relationship? Include the number.

E2. Finish: a shared limb plan supports

E3. Name one thing these evidence sources cannot establish.

L1. What question did your audience ask? How did it change or sharpen your caption?

My next useful step is...

- A. Practise reading the sampled quantity.
- B. Explain shared ancestry with the limb diagram.
- C. Distinguish observation from inference.
- D. Evaluate a new evidence source.